



Reinforcement Learning for Risk Averse Selection

Using Supervised Learning Simulated Environment - Baseball Pitch Selection

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Introduction

Many reinforcement learning **reward functions** maximize the **expected value** in choosing actions. This project looks to **explore differences in reward functions**, and in particular penalize extreme outcomes. This is similar to

- Portfolio Managers worrying about downside volatility in addition to returns
- Navigation algorithms accounting for excess traffic, stoplights, etc.
- Healthcare professionals including severe side effects in patient models

These areas represent meaningful areas of impact but can be notoriously difficult to simulate, so I use **baseball** to explore this spacing. Specifically I use Statcast data to build a simulation environment with supervised learning to learn an RL algorithm for pitch selection, then simulate with models learned from different reward functions.

Supervised Learning Data and Features

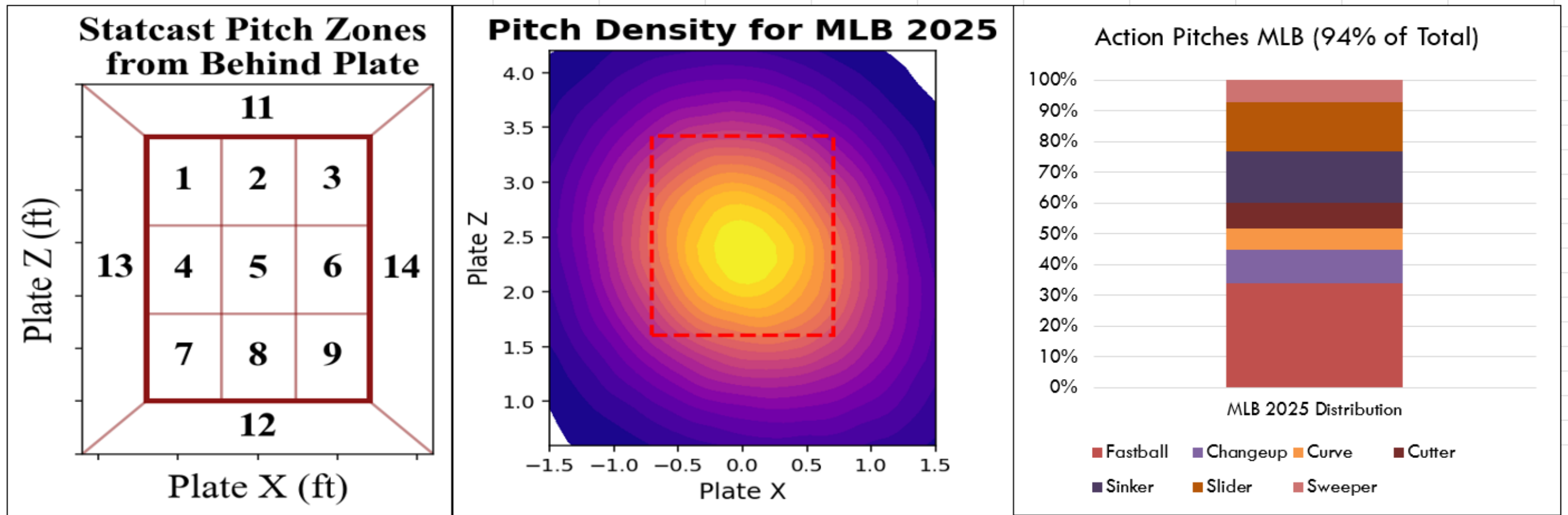
MLB Statcast data contains information on every single pitch as recorded by the TrackMan radar system. Of particular use for this project the fields include:

- Pitch Information and Physics** - the pitch type, speed, break, location, etc.
- Game State** - the count, baserunners, pitcher, batter, fielders, score, teams, etc.
- Batting Physics** - swing length, bat speed, batted ball angle and velocity, etc.

The feature set primarily utilizes the pitch type and physic, batting physics, & generic game state, but individual player information is not included.

Categorical variables are **one-hot encoded**, and **numerical** variables are **scaled to N(0,1)**.

RL States, Actions, Episodes, Epochs, and Reward Functions



Given a **game state** the model selects **pitch type** and **zone of the plate** and generates

- Uniform initial selection** to start with any pitch equally likely
- 162 half inning episodes** appropriate and about what a starter pitches a season
- 20 epoch training cycle** both appropriate and about a 20 year MLB career

The Statcast data was used to train the SL algorithms that perform the statistical simulation.

Reward Functions (4 variants):

- Baseline**: linear reward for outs and runs with minor penalties for other events
- Outs Only**: linear reward only when an out is recorded.
- Runs Only**: linear negative reward whenever a run scores.
- Limit Big Innings**: super-linear penalties for high-run innings.

Supervised Learning Models

Each different dependent variable type warrants use of a different model, with three in total.

- Binary variables** modeled with **Logistic Regression** with an L2 penalty ($\lambda = \frac{1}{2}$)

$$L_{\text{bin}}(w) = - \sum_{i=1}^n [y^{(i)} \log p_i + (1 - y^{(i)}) \log(1 - p_i)] + \lambda \|w\|_2^2, \quad p_i = \sigma(w^\top x^{(i)}) = \frac{1}{1 + e^{-w^\top x^{(i)}}}$$

- Multi-Class variables** modeled with **Softmax Regression** with an L2 penalty ($\lambda = \frac{1}{2}$)

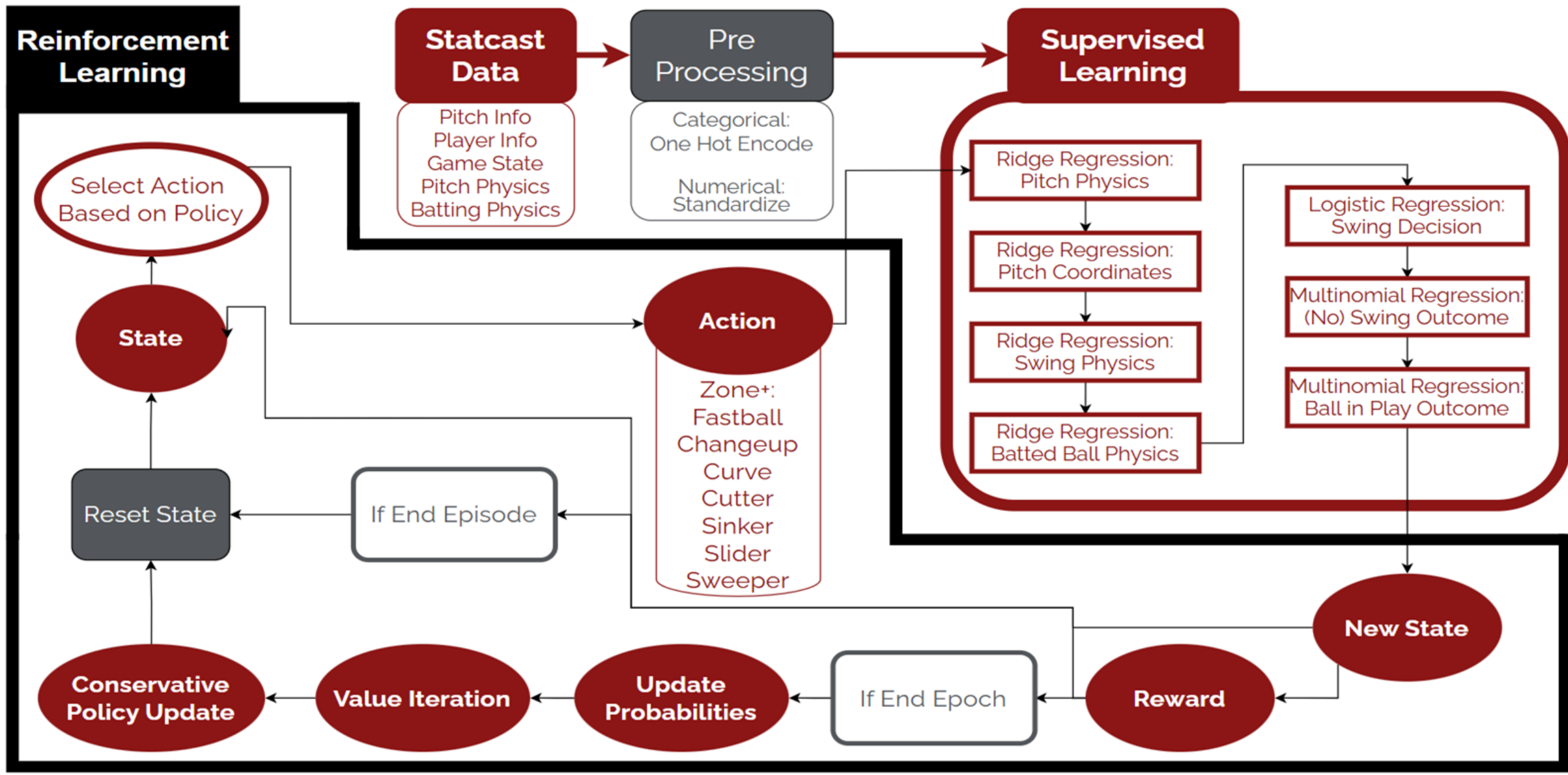
$$L_{\text{multi}}(W) = - \sum_{i=1}^n \log \prod_{l=1}^k p_{il}^{\mathbf{1}\{y^{(i)}=l\}} + \lambda \sum_{l=1}^k \|w_l\|_2^2, \quad p_{il} = \text{softmax}_l(W^\top x^{(i)}) = \frac{\exp(w_l^\top x^{(i)})}{\sum_{j=1}^k \exp(w_j^\top x^{(i)})}$$

- Numerical variables** modeled with **Ridge Regression** (λ selected with CV)

$$L_{\text{ridge}}(w) = \frac{1}{n} \sum_{i=1}^n (y^{(i)} - w^\top x^{(i)})^2 + \lambda \|w\|_2^2.$$

I compare each model to two separate baselines.

- Baseline** the average from the entire training set
- Smart Baseline** the average from the training set given pitch type and zone



Reinforcement Learning Model

Pitch selection is modeled as a finite **Markov Decision Process** ($S, A, \{P_{sa}\}, \gamma, R$) where $s \in S$ are states, $a \in A$ actions, γ discount factor (0.95), and the transition probabilities $\{P_{sa}\}$ and reward R are estimated empirically after each episode.

The value function is computed by **value iteration**

$$V(s) = \max_{a \in A} [R(s, a) + \gamma \sum_{s' \in S} P_{sa}(s', a) V(s')]$$

For each reward function, the model selects a greedy action $a^*(s)$ and updates the optimal policy using **conservative policy iteration** with step size $\alpha = 0.2$ to encourage **exploration**

$$\pi_{\text{new}}(a|s) = (1 - \alpha)\pi_{\text{old}}(a|s) + \alpha \mathbf{1}\{a = a^*(s)\}$$

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Supervised Learning Results and Discussion

Categorical: (Multinomial) Logistic Evaluation Metrics							Numerical: Ridge Regression Evaluation Metrics									
Metric	Set	Model	Swing Decision	No Swing Outcome	Swing Outcome	Hit Into Play	Metric	Release Speed	Vertical Break	Horizontal Break	Plate X	Plate Z	Swing Length	Bat Speed	Launch Angle	Launch Speed
AUPRC	Validate	Supervised	0.82	0.69	0.50	0.23	MSE	9.81	0.142	0.620	0.147	0.216	0.496	26.95	704.78	154.13
		Smart Baseline	0.71	0.63	0.46	0.11		9.96	0.239	0.719	0.148	0.229	0.539	47.86	759.18	190.21
		Baseline	0.48	0.33	0.33	0.10		39.04	1.243	0.801	0.674	0.941	0.686	51.00	826.57	219.32
AUPRC	Train	Supervised	0.82	0.69	0.50	0.23	MSE	8.54	0.126	0.629	0.151	0.212	0.475	26.72	703.33	149.48
		Smart Baseline	0.72	0.63	0.46	0.11		8.67	0.196	0.720	0.152	0.225	0.520	47.22	766.17	184.69
		Baseline	0.48	0.33	0.33	0.10		36.47	1.151	0.810	0.694	0.929	0.661	50.45	830.70	211.80
Log Loss	Validate	Supervised	0.50	0.19	0.99	0.94	r^2	75%	89%	23%	78%	77%	28%	47%	15%	30%
		Smart Baseline	0.59	0.26	1.02	1.17		74%	81%	10%	78%	76%	21%	6%	8%	13%
		Baseline	0.69	0.65	1.08	1.15		0%	0%	0%	0%	0%	0%	0%	0%	0%
Log Loss	Train	Supervised	0.50	0.19	0.99	0.93	r^2	77%	89%	22%	78%	77%	28%	47%	15%	29%
		Smart Baseline	0.59	0.27	1.02	1.13		76%	83%	11%	78%	76%	21%	6%	8%	13%
		Baseline	0.69	0.65	1.08	1.15		0%	0%	0%	0%	0%	0%	0%	0%	0%

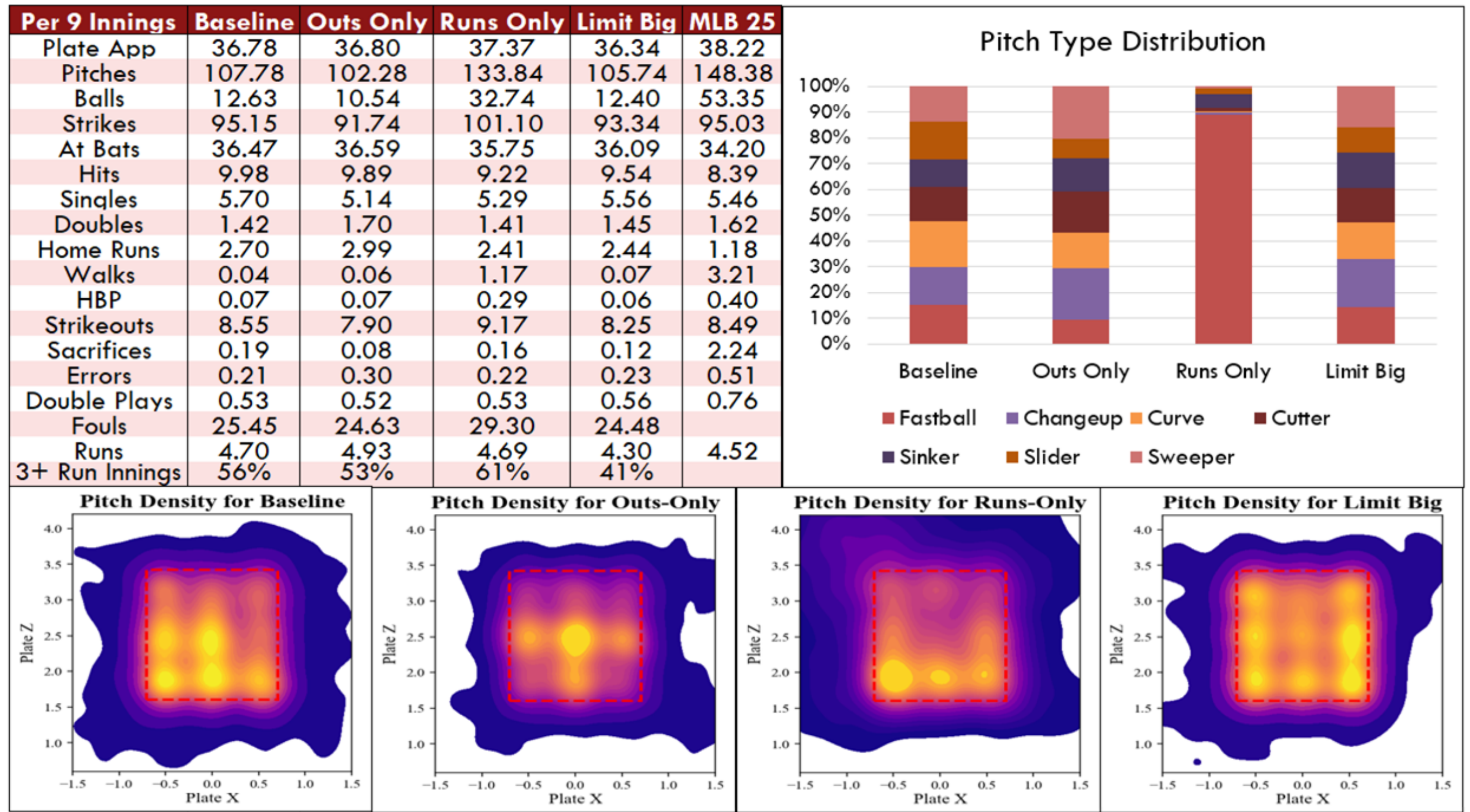
The SL models performed well across training & validation sets with no signs of overfitting

- Categorical Models** outperform in **AUPRC** and **log-loss**
- Numerical Models** outperform baselines; lower **MSE** & higher **r-squared**

SL models outperform baselines in calculating outcome odds, so can be used in simulation.

- AUPRC/Log-Loss SL**: 0.20/1.10, Smart Baseline: 0.13/1.25, Baseline: 0.07/1.54

Reinforcement Learning Results and Discussion



In comparison to the MLB season statistics, the learned models:

- throw significantly fewer balls, so less walks, due to assumed perfect accuracy
 - have more hits and home runs, and only in Limit Big Innings is there less runs
 - throw a similar number of strikes and strikeouts, and have similar other metrics
- All of the different reward function lead to different optimal pitch selection.
- Outs Only** less fastballs, middle of the plate, seeming to maximize balls in play for outs
 - Runs Only** more fastballs, corners of the zone, yielding more walks less multi-base hits
 - Baseline** yielded results right in between outs and runs
 - Limit Big Innings** less 3+ run innings, fewest runs overall, varied pitches and locations

Future

The entire process can be iterated upon and improved by:

- Further data exploration, feature generation, modeling player & ballpark differences
- Continuing to explore SL models and revisiting simplifying assumptions
- Exploration of additional RL algorithms, and more training time
- Improved hyperparameter exploration and tuning across all models